

BRD | Business Requirements Document

Project Name: KRYSTAL Talent Matchmaking Platform

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1. Executive Summary

This document outlines the requirements for the development of **KRYSTAL Talent Matchmaking Platform**, an innovative AI-powered matchmaking platform designed to efficiently connect businesses with talent in the creative industry (models, influencers, actors, digital creators). The primary goal is to democratize access to professional representation opportunities and optimize the talent discovery process, overcoming the limitations of traditional methods and industry fragmentation.

2. Problem

Currently, 95% of creative talent remains completely locked out of professional representation. Geographic, economic, and industry gatekeepers maintain artificial scarcity, limiting access to opportunities. Talent often makes career decisions blindly, without industry expertise or data. On the other hand, traditional talent agencies spend up to 60% of their time on administrative tasks instead of value creation. Talent discovery remains an archaic, relationship-based process, missing massive potential. This \$15 billion+ industry operates at a fraction of its potential.

3. Product Vision

KRYSTAL aims to be the **ONLY** unified talent ecosystem, transforming traditional talent management into an AI-powered extinction event. We envision a platform

where talent has direct access to opportunities and AI-powered career development, and where businesses can discover talent predictively with unprecedented precision.

4. Goals and Objectives

4.1 Business Goals

- Democratize access to opportunities for 95% of unrepresented talent.
- Increase the efficiency of the talent search process, reducing time and costs for businesses.
- Create the leading unified platform in the industry for talent matchmaking.
- Monetize the platform through a subscription model and micro-fees.

4.2 Product MVP Objectives

- Develop and launch a functional matchmaking platform with talent profiles and projects.
- Implement semantic search based on AI embeddings to connect talent and businesses.
- Offer an integrated chat system to facilitate communication between users.
- Validate the business model with a group of beta users before a large-scale launch.

5. Audience / Users

1. Talent: Models, influencers, actors, digital creators, content creators seeking representation, opportunities, and career development.
2. Businesses: Brands, advertising agencies, production studios, casting directors looking for specific talent for their campaigns and projects.

6. Key Features (MVP)

6.1. Talent Functionalities

- **Talent Profile:** Talent can create detailed profiles including demographic data, skills, interests, preferred rates, and availability.

- **Multimedia Portfolio Management:** Talent can upload professional photos and videos to their portfolio.
- **Social Media Integration:** Connect to import key metrics (follower count, engagement rate, etc.) from their public profiles.
- **Jobs / Payments:**
- **MVP Key Feature: AI Agent for Deals:** The AI agent will assist in identifying and suggesting potential jobs for the talent based on their profile and portfolio. It will serve as an initial filter and a recommendation engine.

6.2. Business Functionalities

- **Campaign Administrator:** Businesses can post detailed briefs for their campaigns, including objectives, budget, target audience, desired style, and specific talent requirements.
- **AI Talent Search:** Once a project is published, the AI system will generate a curated list of talent that best fits the project's needs, with a focus on compatibility, ROI potential and *trending status*.
- **Talent Profile Viewing:** Businesses can view the complete profiles and portfolios of AI-suggested talent, but talent will not directly see campaigns or business details until invited to collaborate.

7. Proposed Technical Solution

The platform will be built using a modern and scalable technology stack, with artificial intelligence as its core for precise matchmaking.

- **Frontend (FE):**
 - **Technology:** Next.js (based on React.js)
 - **Details:** Will be used to build the dynamic and responsive user interface for both talent and businesses, leveraging Server-Side Rendering (SSR) capabilities for optimal performance and SEO.
- **Main Backend:**
 - **Technology:** Next.js API Routes (JavaScript/TypeScript)
 - **Details:** Will act as the orchestration layer, handling business logic, user authentication, and serving as a bridge between the frontend, the

database, and the AI microservice.

- Database (DB) & BaaS:
 - Technology: Supabase
 - Details:
 - Database: PostgreSQL, with the `pgvector` extension enabled. This will allow for storing AI-generated vector embeddings directly in the database and performing efficient similarity searches for matchmaking.
 - **Authentication:** Supabase Auth will be used for secure user registration and login.
 - **Document Comment:** For the Proof of Concept (POC), Supabase Auth is a practical and efficient choice. For a production environment, a more scalable architecture would be required. A recommended approach would be to use a robust database like **PostgreSQL** with **Citus** for data management and a specialized vector database like **Pinecone** for handling complex AI workflows and semantic search at scale.
 - Storage: Supabase Storage will manage the efficient storage of large multimedia files (photos, videos) from talent portfolios.
- Artificial Intelligence (AI) Microservice:
 - Technology: Python (utilizing FastAPI for API endpoints).
 - Details: This microservice will be dedicated to heavy data processing and the generation of embeddings, exposing endpoints that will be consumed by the Main Next.js Backend.
 - Specific AI Models to be Used:
 - For Image and Text Embeddings (Multimodal): The CLIP (Contrastive Language-Image Pre-training) model (open-source/Hugging Face versions) will be used. This model will generate 512-dimensional vector embeddings that represent both images from talent portfolios and text descriptions from profiles and project briefs in a common semantic space, enabling precise matchmaking between visual talent content and textual project requirements.
 - For Video Processing: FFmpeg will be utilized for extracting

keyframes from video portfolios. These keyframes will then be individually processed with CLIP to generate their embeddings, which will subsequently be aggregated to represent the entire video.

- For Audio Analysis/Transcription (Optional in MVP, for deeper video content understanding): Pre-trained Speech-to-Text (STT) models (e.g., Wav2Vec2 from Hugging Face) could be used to transcribe audio from videos. The resulting text would then be processed with a text embedding model (such as CLIP or Sentence Transformers like all-MiniLM-L6-v2 which generates 384-dimensional vectors) to capture the verbal essence of the content.
- Social Media Metrics: Numerical metrics (followers, engagement) will be stored in Supabase. The weighting and normalization logic for these metrics, in conjunction with AI embeddings, will be implemented in the Main Backend for the final ranking of suggested talent.

8. Success Metrics

- **Conversation Rate:** The percentage of initiated contacts that lead to a conversation between a business and talent.
- **New vs. Engaged Users:** Tracking the number of new registrations versus the number of users who actively use the platform (e.g., uploading a portfolio, creating a project, sending a message).
- **Deals Transactioned:** The total number of confirmed deals or projects that culminate in a contract through the platform.
- **Monthly Subscribers (Talent & Agencies):** The number of paying users on the platform, which directly validates the subscription model.
- **Match Rate:** Percentage of projects resulting in at least one initiated contact by a business with an AI-suggested talent.
- **Project Conversion Rate:** Percentage of published projects that culminate in a confirmed deal with talent found through the platform.
- **Talent Discovery Time:** Reduction in the average time it takes businesses to find suitable talent.

